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SensOri

A Real-Time Sensory Environment Navigator for Autism-Friendly Outings

How Might We improve travel planning for caregivers of people susceptible to sensory overload by providing early access to objective, real-time sensory environment data?

The Problem

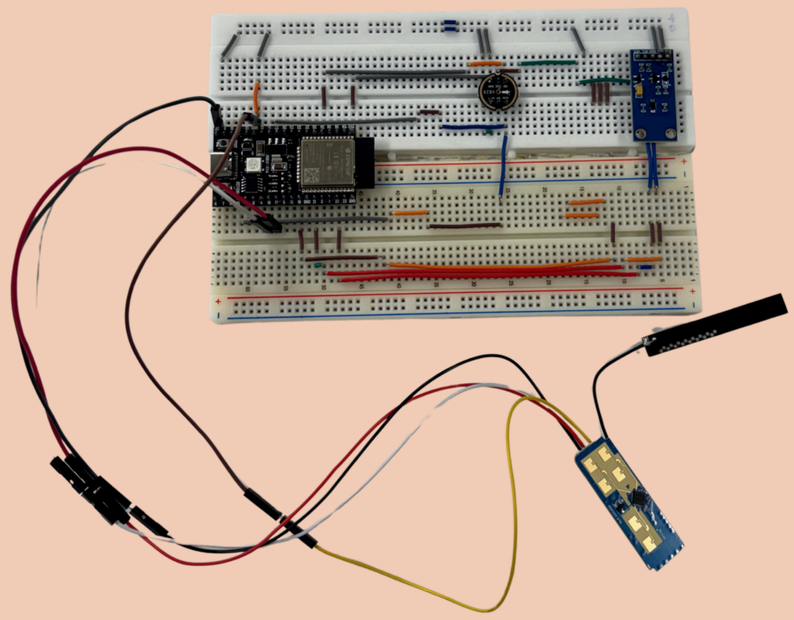
- ~90% of individuals with Autism Spectrum Disorder (ASD) experience sensory sensitivities
- Public spaces (crowds, noise, bright lights) can be overwhelming
- Leads to stress for individuals and unpredictability for caregivers



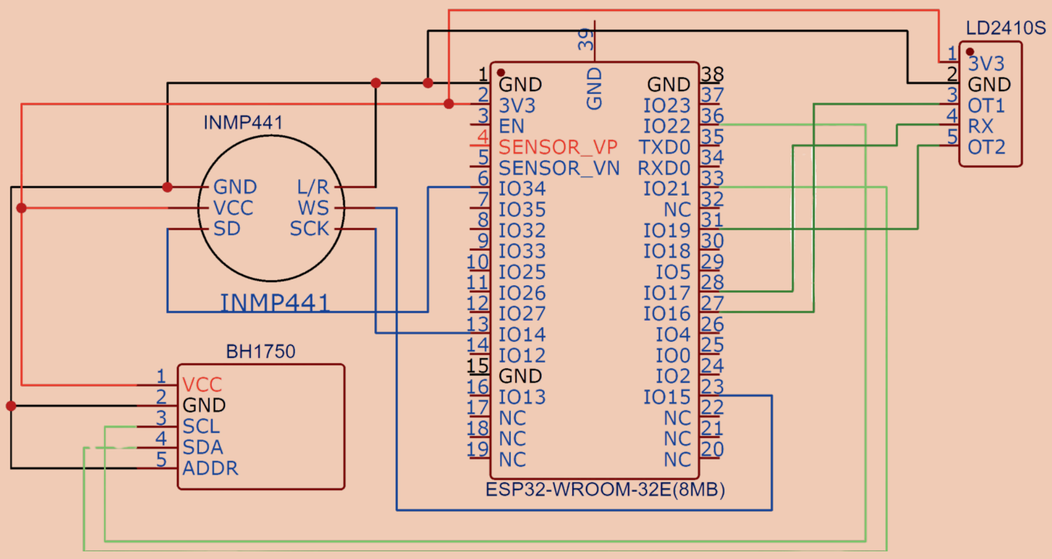
Objectives

- Real-time sensory scores (0–100) for each location
- Predicts future sensory levels using ML
- Uses low-cost IoT sensors in public spaces
- Supports gradual desensitisation with therapy

Circuit Layout



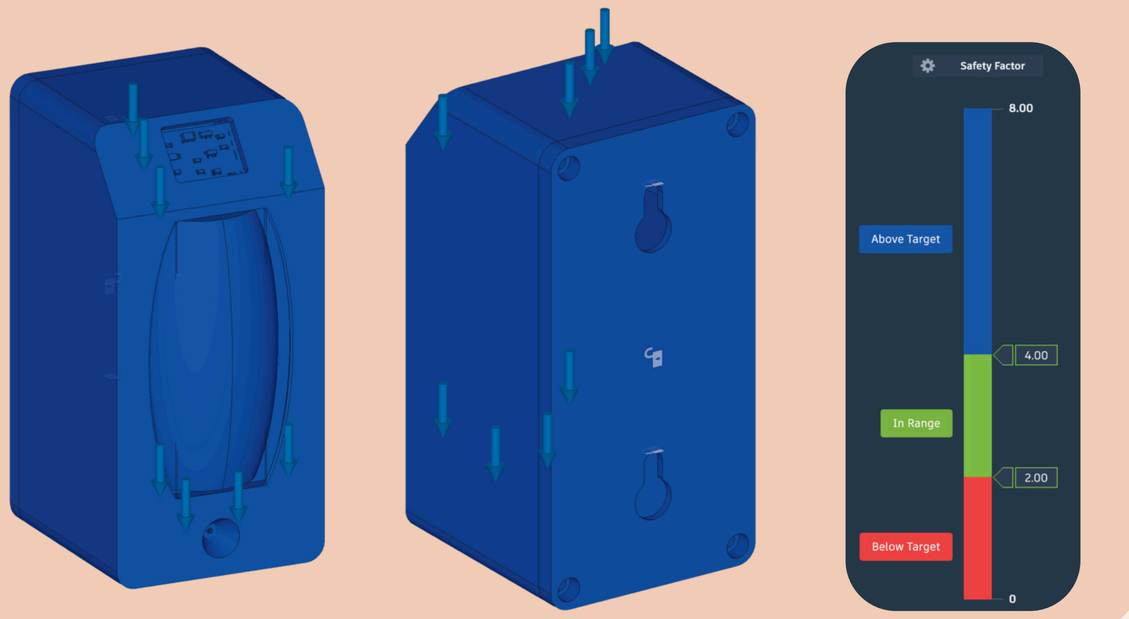
Circuit Schematic



Hardware Mockup



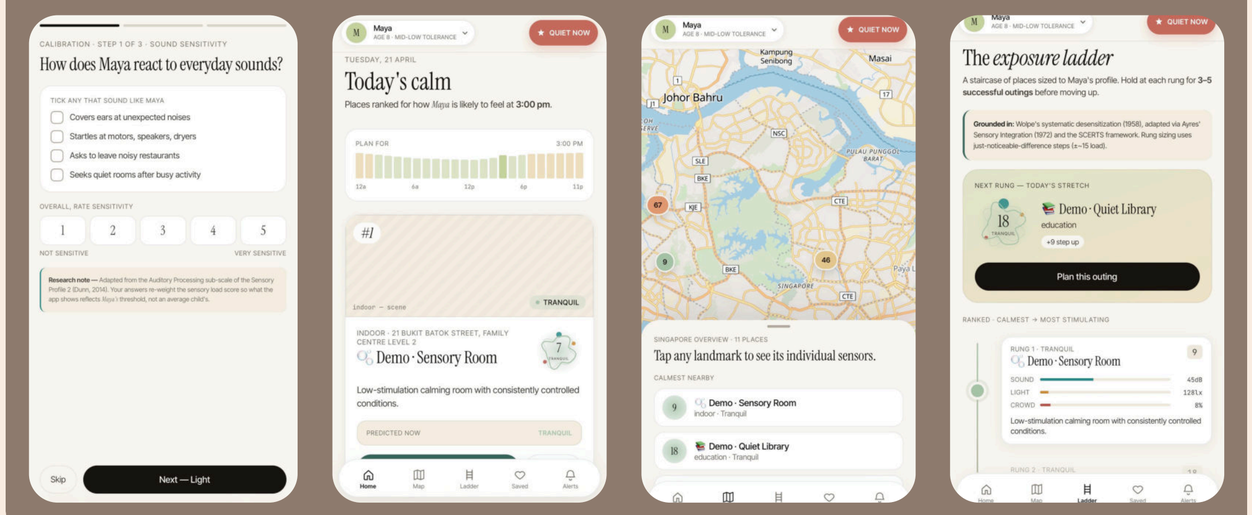
Hardware FEA



CADDE Theory

- Used Adams–Bashforth method to predict future environmental conditions using past sensor data. This is implemented in the app's "Forecast Future Conditions" feature, allowing caregivers to select a date and time and receive a predicted sensory profile.
- Each variable is modelled as a rate of change over time. Since these equations may not have simple solutions, the Adams–Bashforth method helps estimate future values using previous data points.

Software Design (UI)



Electronic Components

- ESP32 reads microphone data using I²S
- INMP441 outputs digital audio (built-in amp & filter)
- BH1750 uses I²C communication
- HLK-LD2450 detects using threshold logic
- Wiring checked to match code and prevent errors

Structure and Materials

- Designed for wall-mounted public use
- Plastic casing for durability and insulation
- Polycarbonate windows for light & sound transmission
- Aluminium bracket for strength and corrosion resistance
- Finite Element Analysis (FEA) was conducted to ensure the casing is durable and structurally stable.